

GADSDEN 19 N, AL, USA

WMO: 722380

Lat: 34.2851N

Lon: 85.9621W

Elev: 1152

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63857

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.5	6.0	7.8	20.8	10.3	9.7	24.5	16.1	37.6	14.7	37.1	4.0	N/A	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	91.6	73.2	89.4	73.2	87.5	73.1	76.9	85.9	75.9	84.8	75.1	83.8	4.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.3	133.6	81.2	73.3	129.2	80.3	72.4	125.4	79.3	41.0	86.0	40.2	85.0	39.3	83.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.9	10.3	DB	11.1	94.8	4.9	3.3	7.6	97.2	4.7	99.2	2.0	101.0	-1.6	103.5	(4)
(5)			WB	9.6	78.6	5.4	1.3	5.7	79.5	2.5	80.3	-0.6	81.0	-4.5	81.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	39.4	43.4	51.4	59.2	66.7	74.4	76.6	76.5	71.4	60.2	49.1	44.3		
	DBStd	15.14	10.29	10.18	9.12	7.31	6.37	3.72	3.53	3.87	5.43	8.27	8.39	9.50		
	HDD50	1060	354	234	95	14	0	0	0	0	0	14	115	234		
	HDD65	3414	794	607	425	197	60	1	0	0	0	193	479	644		
	CDD50	4515	25	48	139	290	518	733	824	822	643	329	88	56		
	CDD65	1394	0	1	4	23	113	284	359	357	207	43	2	1		
	CDH74	11371	0	2	29	155	875	2500	2934	2955	1579	330	12	0		
	CDH80	3962	0	0	1	6	166	912	1113	1180	518	65	1	0		
(14) Wind	WSAvg	4.0	5.2	5.6	5.4	4.5	3.0	2.7	2.8	2.9	3.0	3.8	4.5	5.2		
(15) Precipitation	PrecAvg	55.7	5.6	5.1	5.9	4.4	4.7	3.9	4.3	3.7	4.2	3.1	4.5	5.3		
	PrecMax	65.7	10.0	11.5	10.4	7.6	13.3	9.1	8.0	8.0	8.6	8.3	10.7	9.3		
	PrecMin	36.9	2.4	2.3	1.8	2.4	0.7	1.1	1.7	1.1	0.4	0.0	2.1	2.6		
	PrecStd	8.1	1.7	2.2	2.0	1.4	2.8	2.2	1.7	1.6	2.4	2.4	2.1	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	72.1	77.5	80.7	87.1	92.9	93.6	96.0	91.5	85.9	75.5	69.2		
		MCWB	59.1	62.3	62.5	64.5	70.6	72.9	73.9	75.4	69.9	67.5	60.5	62.9		
	2%	DB	62.6	67.1	73.5	78.0	84.0	89.8	90.6	92.3	88.6	81.1	69.9	66.3		
		MCWB	57.2	60.5	61.0	63.7	69.9	73.2	74.6	73.9	70.5	66.3	58.9	61.8		
	5%	DB	59.0	63.7	70.2	75.7	81.7	87.8	88.5	89.2	86.0	77.6	66.7	63.0		
		MCWB	53.6	57.1	58.7	63.0	69.2	73.1	74.7	73.6	70.7	64.3	58.4	58.9		
	10%	DB	55.5	60.0	66.4	72.8	79.5	85.7	86.4	86.6	82.9	74.1	63.9	59.7		
		MCWB	50.8	54.3	56.8	61.5	67.8	73.1	74.2	73.3	70.5	62.6	56.7	56.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.6	64.3	64.6	68.4	73.2	77.4	78.2	77.9	75.3	72.1	67.3	66.5		
		MCDB	64.7	69.4	71.2	76.2	81.9	86.1	86.5	87.4	83.1	82.1	68.2	68.0		
	2%	WB	58.7	62.5	63.0	66.8	71.8	76.1	76.9	76.7	74.1	70.1	64.0	63.5		
		MCDB	60.5	66.8	70.0	74.2	80.6	85.4	85.9	85.7	82.7	77.5	66.7	66.2		
	5%	WB	54.9	60.2	61.1	65.2	70.7	75.1	76.0	75.5	73.1	68.0	60.8	59.7		
		MCDB	57.0	63.9	67.0	72.3	79.2	84.6	85.1	83.8	81.7	74.1	63.7	62.2		
	10%	WB	50.6	56.3	59.1	63.7	69.6	74.0	75.2	74.6	71.8	65.4	57.3	56.2		
		MCDB	53.6	59.2	64.7	70.7	77.6	83.2	84.0	82.3	79.3	70.8	62.0	59.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.3	19.6	20.9	20.1	20.5	18.3	19.1	20.1	21.3	21.4	16.8		
		MCDBR	19.8	20.6	22.4	22.9	22.6	23.9	21.6	23.1	24.3	24.2	23.6	17.8		
		MCWBR	16.0	14.1	12.6	12.1	10.8	9.7	8.4	8.7	9.4	11.7	14.7	14.4		
	5% WB	MCDBR	15.3	16.2	18.1	19.2	18.8	20.2	18.8	18.3	18.4	19.4	16.1	16.5		
		MCWBR	14.6	12.9	11.4	11.2	9.8	10.0	8.1	8.0	8.0	11.1	13.4	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.314	0.353	0.383	0.437	0.470	0.519	0.517	0.418	0.367	0.333	0.312		
	taud		2.519	2.506	2.419	2.318	2.210	2.154	2.035	2.028	2.295	2.428	2.466	2.531		
	Ebn at Noon		286	298	292	286	271	261	247	245	267	274	274	279		
	Edh at Noon		26	30	35	41	46	48	54	54	39	32	27	25		
(44) All-Sky Solar Radiation	RadAvg		800	966	1326	1718	1882	1965	1874	1742	1507	1242	907	681		
	RadStd		58	124	122	115	149	152	119	95	138	159	101	67		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+275	+203

Nomenclature: See separate page